

[EA&B] CDXR: A Contract-Grounded Evaluation Architecture for Tabular Leakage Repair

Author Names Pending
Affiliations Pending
City Pending, Country Pending
author-email-pending@example.org

Abstract

Leakage repair claims often conflate four objects: whether information is valid at prediction time, whether a blind policy can localize it, whether a learner exploits it, and whether deletion restores a boundary-valid evaluation. We introduce CDXR, a contract-grounded evaluation architecture that separates construction validity (C), detectability (D), exploitability (X), and repair (R) through explicit information, policy-access, cost, and falsification interfaces. LeakBench-Tab instantiates CDXR with 27,500 paired measurement cells and a 803,000-cell Full-B1 governance execution over 5,500 keys, five policies, two cost contracts, three budgets, and 20 matched-random seeds. Under the primary semantic-group 20% contract, blind mutual-information removal (P3) improves directional repair over random removal by +0.095 (95% task-cluster CI [+0.079,+0.110]) and complete leakage-group recall by +0.483 ([+0.462,+0.503]). Yet it also increases overcorrection by +0.052 ([+0.023,+0.085]), while its legacy score-distance interval crosses zero (+0.043, [-0.005,+0.084]). Three additional blind policies exhibit the same localization-overcorrection tension, so the joint gate classifies all four as tradeoffs rather than successful repairs. A separately frozen LR/RF/LightGBM amendment finds no reliable learner interaction for the narrower encoded-column P3 score contrast, but all learners fail the overcorrection gate. CDXR thus contributes more than a benchmark ranking: it specifies when a repair claim is admissible and can falsify apparently positive score recovery when semantic and intervention outcomes disagree.

Keywords

data leakage, evaluation architecture, tabular benchmarks, feature governance, reproducible experiments

1 Introduction

Tabular prediction pipelines routinely inherit fields whose values are only available after the prediction event, are derived from the target, or summarize future entities and outcomes. Such contamination can invalidate evaluation and distort model comparison even when the learning algorithm and held-out split are otherwise conventional [8, 9]. The practical response is often to remove suspicious fields, but this response combines three different questions: whether a field is semantically valid at prediction time, whether an oracle-blind statistic can locate it, and whether a learner exploits it enough to change measured performance.

Existing leakage studies establish the importance of prediction boundaries, scientific validity, and domain-specific label leakage [4, 14]. Detection work also spans code analysis and empirical contamination audits [11, 16]. These perspectives do not

by themselves determine whether a deployable, budget-limited removal policy will repair an affected tabular pipeline. A detector can rank target proxies highly yet fail on future aggregation or entity structure. Conversely, a semantically invalid field can be hard to rank but still inflate a learner’s score.

This gap matters because leakage response is an intervention, not merely a diagnosis. Removing a field changes the representation on which the learner is fitted and may discard legitimate predictive information along with the contamination. Comparing a removal policy only with the unmodified table confounds two effects: locating invalid information and reducing model capacity. Comparing it only with an oracle mask instead asks whether the policy is perfect, not whether its ranking is useful under realistic constraints. A matched random policy supplies the missing negative control: it estimates how much movement toward a strict reference is obtained merely by deleting the same number of encoded fields.

A credible benchmark for this question must preserve several separations at once. Semantic invalidity must be declared before examining feature scores; diagnostics must be fitted without the contamination mask; strict, permissive, and governed learners must share task content and test rows; and uncertainty must respect the task hierarchy rather than treating repeated mechanisms, strengths, seeds, or models as independent datasets. These requirements turn a seemingly simple feature-ranking comparison into a protocol-design problem.

We address this protocol-design problem with CDXR, an evaluation architecture rather than a new selector. CDXR makes four interfaces explicit. An information contract freezes prediction-time validity; a policy contract enumerates exactly which training data and metadata a selector may access; a cost contract maps semantic groups to charged intervention units; and an evaluation contract computes an oracle-isolated repair vector after selection. The oracle mask is therefore neither a training label nor hidden tuning input. It is an evaluation object that can falsify a repair claim. Lineage-aware, time-aware, and group-aware policies can be attached to the same interface by declaring additional policy-visible metadata without changing the evaluator.

LeakBench-Tab is the controlled implementation of this architecture. It crosses eleven audited mechanisms with tasks, strengths, injection seeds, and learners for C/D/X measurement, then executes Full-B1 over four learned policies, 20 matched-random realizations, two cost units, and three budgets for R. Our analysis asks four questions. **RQ1** characterizes how D and X vary across mechanisms. **RQ2** asks whether score recovery is accompanied by directional and semantic repair rather than overcorrection. **RQ3** tests policy and cost-contract sensitivity after integrating random-policy variance. **RQ4** audits whether the narrower P3 score contrast interacts with the downstream learner.

Contributions. The paper makes four bounded contributions. First, it defines CDXR as an executable architecture with typed inputs, policy-visible information, cost maps, output vectors, and an explicit unsupported-claim gate. Second, it provides a frozen factorial registry that varies task archetype, contamination mechanism, strength, seed, and learner while preserving paired strict and full evaluation. Third, Full-B1 evaluates four blind policies against 20-seed matched random removal under semantic-group and encoded-column cost, producing 803,000 validated downstream cells. Fourth, the architecture changes the scientific conclusion: learned policies improve localization and directional movement, but their increased overcorrection prevents a general repair claim. The artifact binds every paper-facings estimate to canonical ledgers, protocol manifests, analysis code, and content hashes.

2 CDXR Evaluation Architecture

2.1 Prediction Boundary and the C/D/X Axes

Let $q = (t, m, a, s)$ index a controlled cell by task t , contamination mechanism m , strength a , and injection seed s . Its encoded feature universe is $\mathcal{F}_q$, and the construction contract identifies the contaminated subset $\mathcal{L}_q \subseteq \mathcal{F}_q$. The strict and permissive views are therefore

$$\mathcal{F}_q^{\text{strict}} = \mathcal{F}_q \setminus \mathcal{L}_q, \quad \mathcal{F}_q^{\text{full}} = \mathcal{F}_q. \quad (1)$$

Each task declares an information boundary at prediction time t_p . For encoded field j , construction validity is

$$C_{qj} = \mathbf{1}[j \in \mathcal{L}_q] = \mathbf{1}[j \text{ is unavailable or target-derived at } t_p]. \quad (2)$$

C_{qj} is fixed by task semantics and mechanism construction. It is not inferred from association, attribution, or downstream performance.

An oracle-blind diagnostic assigns score s_{qj} without reading C_{qj} or the test rows. We measure localization with average precision and normalize it by the contaminated field prevalence π :

$$\pi_q = \frac{|\mathcal{L}_q|}{|\mathcal{F}_q|}, \quad D_q = \frac{AP_q - \pi_q}{1 - \pi_q}. \quad (3)$$

The primary diagnostic is mutual information computed on training rows with a fixed random state. D is conditional on the diagnostic and representation.

For learner ℓ , exploitability is the paired AUROC difference

$$X_{q\ell} = A_{q\ell}^{\text{full}} - A_{q\ell}^{\text{strict}}, \quad (4)$$

where ℓ indexes the downstream learner and A denotes test AUROC. The strict view removes construction-labeled contaminated fields; the permissive view retains them. Positive X measures performance inflation under this protocol. An interval containing zero is reported as an imprecise or unsupported directional effect, not as evidence of equivalence.

CDXR completes these measurements with repair outcome R . Let an information contract be

$$\mathcal{I}_q = (t_p, \mathcal{F}_q, \mathcal{G}_q, \mathcal{L}_q, \rho_q), \quad (5)$$

where t_p is the prediction event, $\mathcal{G}_q$ partitions encoded columns into semantic groups, and ρ_q records availability or lineage evidence. A policy contract declares an access set

$$\mathcal{A}_r \subseteq \{X^{\text{train}}, y^{\text{train}}, \mathcal{G}, t, \text{entity}, \rho\} \quad (6)$$

and must satisfy

$$\mathcal{L}_q \notin \mathcal{A}_r \quad \text{for every non-oracle policy } r. \quad (7)$$

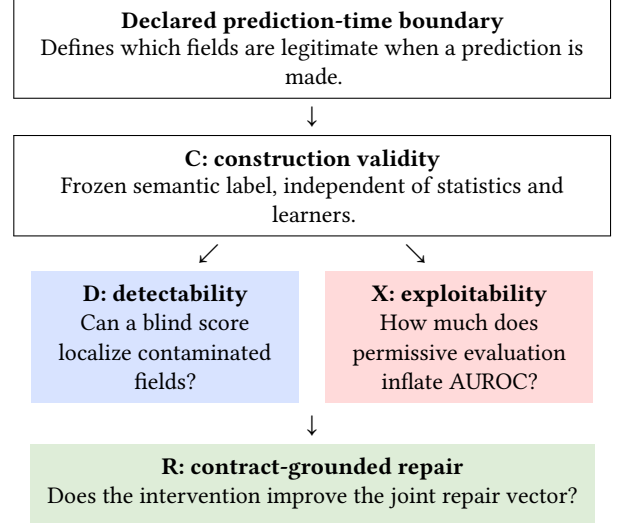


Figure 1: CDXR evaluation architecture. The information contract fixes C; a declared policy-access interface produces D-conditioned selections; learners produce X; and an oracle-isolated evaluator computes R under an explicit cost map. A claim is unsupported when required repair coordinates conflict.

Thus a group-aware selector may read $\mathcal{G}_q$, and a time-aware selector may read training timestamps, but neither may read the held-back invalidity mask.

2.2 Governance Objective

For policy r , cost contract c , and budget b , let $\mathcal{R}_{q,r,c,b} \subseteq \mathcal{F}_q$ be its selected removal set and $\kappa_{q,c}(R)$ its charged cost. Encoded-column cost counts columns; semantic cost counts every touched group once:

$$\kappa_{q,\text{col}}(R) = |R|, \quad \kappa_{q,\text{grp}}(R) = |\{g \in \mathcal{G}_q : g \cap R \neq \emptyset\}|. \quad (8)$$

The governed view and matched-cost constraint are

$$\mathcal{F}_{q,r,c,b}^{\text{gov}} = \mathcal{F}_q \setminus \mathcal{R}_{q,r,c,b}, \quad \kappa_{q,c}(\mathcal{R}_{q,r,c,b}) = b. \quad (9)$$

Let A_s , A_f , and A_g denote strict, full, and governed test AUROC for a fixed q and learner. We define strict distance reduction (SDR) as

$$\text{SDR}_{q,r,k} = |A_{f,q} - A_{s,q}| - |A_{g,q,r,k} - A_{s,q}|. \quad (10)$$

Positive SDR means governance moves the evaluated learner closer to the strict reference. This symmetric distance avoids treating negative residual harm as automatic success. We also report contaminated field recall and legitimate retention:

$$\text{Rec}_{q,r,k} = \frac{|\mathcal{R}_{q,r,k} \cap \mathcal{L}_q|}{|\mathcal{L}_q|}, \quad (11)$$

$$\text{Ret}_{q,r,k} = 1 - \frac{|\mathcal{R}_{q,r,k} \setminus \mathcal{L}_q|}{|\mathcal{F}_q \setminus \mathcal{L}_q|}. \quad (12)$$

These outcomes are evaluation quantities and can read C after a policy has selected fields; non-oracle policy selection cannot.

SDR alone cannot distinguish correct repair from crossing past the strict reference. With $h_q = A_{f,q} - A_{s,q}$ and $z_{q,r} =$

$A_{g,q,r} - A_{s,q}$, define

$$\text{Dir}_{q,r} = |h_q| - \max\{\text{sgn}(h_q)z_{q,r}, 0\}, \quad (13)$$

$$\text{Over}_{q,r} = \max\{-\text{sgn}(h_q)z_{q,r}, 0\}. \quad (14)$$

For semantic groups $\mathcal{H}_q \subseteq \mathcal{G}_q$ labeled invalid, complete-group recall is

$$\text{GRec}_{q,r} = \frac{1}{|\mathcal{H}_q|} \sum_{g \in \mathcal{H}_q} \mathbf{1}[g \subseteq \mathcal{R}_{q,r}]. \quad (15)$$

CDXR returns the repair vector

$$R_{q,r,c,b} = (\text{SDR}, \text{Dir}, \text{GRec}, \text{Rec}, \text{Ret}, \text{Over}, \text{Dist}_0). \quad (16)$$

A score-recovery claim is *unsupported* when its required SDR interval crosses zero. A semantic-repair claim is also unsupported when localization or direction fails, or when the policy improves them only by reliably increasing overcorrection or zero-opportunity distortion. This falsification rule is the architectural difference between CDXR and a renamed metric taxonomy.

2.3 Evaluation Conditions

Four conditions distinguish the benchmark from ordinary feature selection. *Semantic precedence* requires $\mathcal{L}_q$ to be determined by the prediction-time contract rather than by a score threshold. A feature can be invalid with low marginal association, or valid with high association; neither case is contradictory. *Oracle isolation* requires the selection map $\mathcal{R}_{q,r,k} = \phi_r(X_q^{\text{train}}, y_q^{\text{train}}, k)$ to exclude $\mathcal{L}_q$ for non-oracle policies. The mask enters only after $\mathcal{R}_{q,r,k}$ is frozen, when recall and retention are evaluated.

Cost matching holds $\kappa_{q,c}$ fixed within each policy comparison. One-hot expansion makes a semantic field and an encoded column different intervention units: the eight M09 source indicators cost eight under $c = \text{col}$ and one under $c = \text{grp}$. Full-B1 executes both contracts rather than retrospectively recomposing one from the other. The semantic-group contract is primary for repair validity; encoded-column cost is the representation sensitivity.

Paired reference evaluation keeps the task, split, mechanism, strength, injection seed, and learner seed fixed when strict, full, and governed AUROC are compared. SDR is bounded by $[-1, 1]$ because it is a difference of two AUROC distances. Positive SDR indicates movement toward the strict score, but not recovery of the strict feature set: a policy may reach a similar AUROC after removing the wrong fields. Recall and retention are reported alongside SDR precisely to reveal that ambiguity. Likewise, negative SDR identifies movement away from the strict reference but does not by itself distinguish residual contamination from over-removal of legitimate signal.

These conditions define the scope of a governance claim. A policy is useful in this benchmark only when it improves the paired distance under equal removal cost and its localization trade-off remains visible. The formulation does not assert that the strict model is optimal for a business objective, only that it is the evaluation reference consistent with the declared information boundary.

2.4 Composable Policies and Claim Admission

A CDXR policy plug-in is the tuple

$$\Pi_r = (\mathcal{A}_r, \sigma_r, \tau_r, \kappa_c), \quad (17)$$

where σ_r maps accessible training information to column or group scores, τ_r is a deterministic tie and truncation rule, and κ_c is the charged cost map. The selector emits only a selection hash and the removed group/column identifiers. A separate evaluator joins that immutable output to $\mathcal{L}_q$ and test predictions. This interface has two practical consequences. First, a new policy can consume lineage or temporal metadata by expanding $\mathcal{A}_r$ without modifying the oracle evaluator. Second, two policies cannot be compared unless their κ_c , budget, downstream learner, split, and task key agree.

Let $L(u)$ and $U(u)$ denote the lower and upper limits of a task-cluster interval for repair-coordinate contrast u . The frozen semantic-corroboration gate is

$$\text{Sem}(r) = \mathbf{1}[L(\Delta\text{Dir}) > 0 \wedge L(\Delta\text{GRec}) > 0 \wedge \overline{\Delta\text{Over}} \leq 0 \wedge \overline{\Delta\text{Ret}} \geq 0 \wedge \overline{\Delta\text{Dist}_0} \leq 0]. \quad (18)$$

If localization is confirmed but an adverse coordinate is also confirmed, the status is `TRADEOFF`. Localization without downstream corroboration is `LOCALIZATION-ONLY`; positive SDR alone is `SCORE-RECOVERY-ONLY`; otherwise the result is negative or not evaluable. The precedence is fixed before effect inspection.

CDXR rejects a repair claim before statistics when its prediction event is undeclared, a non-oracle selector can read $\mathcal{L}_q$, intervention costs are unmatched, or strict/full/governed evaluations do not share the same task key. After statistics, it rejects wording stronger than the claim-gate status. This separation between protocol admission and empirical corroboration makes the architecture usable as both a benchmark harness and a reviewer-facing validity check.

3 Benchmark and Experimental Design

3.1 Controlled Registry

The controlled registry contains 20 deterministic panel tasks spanning five data-generating archetypes. Each task has a chronological 60/20/20 split and separates generator seeds from injection seeds. Eleven mechanisms cover direct target copies (M01), noisy proxies (M02), nonlinear transforms (M03), post-outcome aggregates (M04), temporal look-ahead (M05), redundant clusters (M06), sparse subgroup leakage (M07), entity leakage (M08), source leakage (M09), mixed legitimate and contaminated components (M10), and graph-mediated leakage (M11). Categories are assigned by construction: M01, M02, M06, and M10 are simple; M03, M07, and M11 are boundary mechanisms; M04, M05, M08, and M09 are structured. Categories are not revised after observing results.

The five base archetypes are linear, interaction, nonlinear, sparse, and drifting binary prediction tasks. Each contains entity and time structure and between 12 and 24 original numeric fields. Clean signal combines row-level covariates with entity effects and a periodic time component; the archetype then changes the functional form rather than the evaluation contract. Four tasks instantiate each archetype. Task generator seeds are deterministic functions of the panel identifier, so changing an injection or learner seed cannot regenerate the clean table. All tasks are ordered by timestamp before the 60/20/20 split, and injection is verified not to alter the frozen row indices.

This panel is designed for controlled variation rather than realism by volume. It supplies repeated task structure in which semantic contracts and injected features are exactly auditable. The five natural cases introduced below serve a different purpose:

they test whether boundary-specific contamination remains visible in selected real tables, but they do not replace the controlled panel as the inferential unit.

The core matrix crosses 20 tasks, 11 mechanisms, five strengths, five injection seeds, and five learners, yielding 27,500 paired cells. Logistic regression is the linear baseline. The remaining learners are random forest [1], CatBoost [13], LightGBM [10], and TabM [6]. Preprocessing and diagnostic fitting use training data only. The design spans linear, bagged-tree, boosted-tree, and neural tabular families without treating family differences as causal capacity effects.

3.2 Mechanism Families and Strength Semantics

The registry varies how invalid information becomes available to a learner. The *simple* family contains fields whose encoded columns directly expose or redundantly repeat target information. M01 copies the current label with a strength-dependent flip rate; M02 adds noise to a centered target proxy; M06 creates a correlated cluster whose size increases across strength levels; and M10 pairs one legitimate copy of an original field with one contaminated target proxy. M10 is deliberately mixed: the strict view removes only its contaminated component and retains the legitimate component, preventing the oracle reference from benefiting merely by deleting an entire injected group.

The *structured* family makes validity depend on relations not represented by a single contemporaneous field. M04 aggregates labels from a strictly future window; M05 introduces a configurable look-ahead offset; and M08 computes a shrunken, strictly future label rate within entity. All three exclude the current row's own label. M09 assigns one of eight source categories with outcome-dependent source probabilities and represents the source as a complete one-hot block. It does not use target-rate encoding, but the source identity is deployment-invalid under the declared construction. Consequently, its detectability and removal cost are representation-conditional.

The *boundary* family stresses cases between those poles. M03 multiplies a target-derived signal by the side of a threshold on a legitimate field, hiding marginal association behind an interaction. M07 exposes a target proxy only within a sampled subgroup, with subgroup prevalence varying by strength. M11 distributes target-derived signal across a connected component whose size also varies with strength. These mechanisms are not claimed to enumerate all forms of leakage. They are controlled probes for directness, sparsity, interaction, redundancy, time, entity, source, and group structure.

Strength labels S1–S5 correspond to values 0.2, 0.4, 0.6, 0.8, and 1.0, but the operational parameter is mechanism-specific. Strength scales signal for direct proxies, changes subgroup coverage for M07, and changes group size for M06 and M11. It therefore defines an ordered intervention within a mechanism, not a common physical scale across mechanisms. Cross-mechanism comparisons average the frozen five-level profile; they do not interpret S3 in one mechanism as quantitatively equivalent to S3 in another.

3.3 Diagnostics and Learner Protocol

The primary diagnostic assigns each encoded field a univariate MI score using training rows and a fixed random state. Localization is evaluated after scoring with normalized average precision, which adjusts for the mechanism-specific prevalence

of contaminated encoded columns. Average precision is appropriate for this imbalanced ranking problem because most fields are legitimate and the relevant operating region is concentrated near the top of the ranking [15]. We also record top-five contaminated-field recall as an operational summary, but the normalized AP metric supplies D in the C/D/X analysis.

A frozen sensitivity suite evaluates three alternatives: absolute Pearson correlation on training rows, absolute coefficients from standardized logistic regression, and validation-set permutation importance from a training-fitted random forest. None receives the contamination mask before producing scores. The four-method suite contains 22,000 successful diagnostic cells. Because the protocol did not freeze simultaneous paired method-comparison intervals, these results diagnose method dependence but do not support an inferential ranking of detectors. In particular, a retrospectively selected best method is an optimistic labeled-benchmark summary, not a deployable selector.

The downstream panel fixes five named implementations. LR uses standardized features and logistic regression; RF uses 250 trees; CatBoost and LightGBM use validation-only early stopping; and TabM uses the official pinned package with 32 ensemble members, training-fitted standardization, deterministic execution, and validation-loss checkpointing. Test labels are used only to compute final AUROC. Missing packages or accelerators fail a cell rather than triggering a fallback under another model identity. The completed canonical matrix contains all 27,500 planned model cells, so no failed-cell exclusion enters the reported means.

3.4 Matched-Cost Governance Policies

The governance experiment uses the same 5,500 combinations of task, mechanism, strength, and seed. Full-B1 uses logistic regression downstream so policy and cost effects are not confounded with a learner expansion. P0 retains all fields, and P1 removes the construction-labeled mask; both are analysis-time references. P2 samples uniformly under 20 frozen governance seeds. P3 ranks by training-side mutual information, P4 by absolute point-biserial correlation, P5 by the absolute coefficient of a standardized logistic model, and P6 by three-fold cross-fitted RF permutation importance. P2–P6 receive only the declared training view and opaque group membership. If $\omega \in \Omega$ denotes one random realization and $s_{q,r}(j)$ is the training-only score for deterministic policy r , then

$$\mathcal{R}_{q,P2,c,b}^{(\omega)} \sim \text{Unif}\{R : \kappa_{q,c}(R) = b\}, \quad (19)$$

$$\mathcal{R}_{q,r,c,b} = \text{Top}_{\kappa_{q,c}=b}\{s_{q,r}(j) : j \in \mathcal{F}_q\}, \quad r \in \{P3:P6\}. \quad (20)$$

For nominal fractions $p \in \{5\%, 10\%, 20\%\}$ and cost-universe size $d_{q,c}$, every policy uses

$$b_{q,c}(p) = \max(1, \text{round}(pd_{q,c})). \quad (21)$$

Thus the comparison satisfies the cellwise cost identity

$$\kappa_{q,c}(\mathcal{R}_{q,P2}^{(\omega)}) = \kappa_{q,c}(\mathcal{R}_{q,r}) = b_{q,c}(p). \quad (22)$$

For every repair coordinate u , the paired estimand is

$$\Delta_{q,r,c,p}^{(u)} = R_{q,r,c,p}^{(u)} - \frac{1}{|\Omega|} \sum_{\omega \in \Omega} R_{q,P2,c,p}^{(u,\omega)}. \quad (23)$$

P2 variance is integrated inside each key before task resampling.

The clean runner loads frozen bundles, verifies bundle and split hashes, and does not inject, split, or mutate data. Non-oracle

selectors never receive the contamination mask. Strict and full baselines are shared only within an identical task, mechanism, strength, and seed key. All 792,000 governed rows and 11,000 baselines completed across 5,500 keys, with zero failures or duplicate run identifiers. A separately frozen encoded-column 20% amendment repeats only P3 versus P2 for LR, RF, and LightGBM; it answers learner interaction for that narrow score contrast and is not merged into Full-B1.

3.5 Natural-Case Audit

Five selected public or locally mirrored tables instantiate concrete prediction boundaries: call completion for Bank Marketing, loan origination for Lending Club, scheduled departure for U.S. flights, inspection completion for Chicago food inspections, and complaint creation for NYC 311. For each task, the strict view removes fields declared unavailable at that boundary and the permissive view retains them. The harm audit crosses four CPU learner families with three seeds, producing 60 paired cells. A separate LR governance sensitivity uses the 20% retained-feature budget, three training seeds, and 20 P2 governance seeds, producing 315 policy cells. The five cases are selected rather than sampled, so both analyses remain descriptive fixed-case evidence.

Categorical preprocessing is fitted on training rows. Categories absent from training map to a reserved unknown value, missing categories receive a separate value, and date strings that had previously been globally encoded are dropped. This policy prevents validation or test vocabularies from defining the training representation. The case selection is intentionally heterogeneous in table width and contamination count, but it is not random and does not define a population of application datasets. One Lending Club source remains a local mirror whose redistribution license is unresolved; the paper reports its boundary-specific result but excludes the raw file from the public artifact.

3.6 Freeze and Amendment Discipline

Confirmatory execution is separated from protocol development. The canonical manifest binds task bundles, split hashes, configuration, runner code, model identity, and source result tables. Pilot namespaces use disjoint task seeds. Once a confirmatory protocol is frozen, a change requires an explicit amendment and replacement of the affected cells rather than silent in-place repair.

The canonical table incorporates four documented corrections. M04, M05, and M08 use strictly future definitions that satisfy their construction contracts. M10's strict mask retains its legitimate injected field. MI uses a fixed random state on training data for the affected diagnostic cells, and natural-case preprocessing is training-fitted. Superseded rows are retained for provenance but excluded from canonical claim generation. These amendments narrow the evidence to the final protocol and prevent earlier, semantically different outputs from being averaged into the paper results.

The RF/LightGBM learner study is disclosed as a reviewer-motivated post-run amendment rather than presented as pre-specified Full-B1 scope. Its learner factories, hyperparameters, seeds, and encoded-column 20% estimand were frozen before those outcomes were inspected. The amendment re-fits strict and full baselines with the same factories instead of reusing earlier canonical fits, preserves within-run pairing, and remains in a

separate namespace. This timing permits a bounded interaction sensitivity but not a claim that the three learners or reported metrics were selected ex ante.

3.7 Statistical Analysis and Claim Scope

The controlled task is the independent unit. Core measurement intervals use a 20,000-draw hierarchical bootstrap that samples tasks and then injection seeds. Mechanism tests use task-level sign flips with Holm correction over the eleven mechanisms. The three category contrasts form a separate Holm family. The pre-specified simple-minus-structured contrast receives the core binary support decision; individual mechanism profiles remain descriptive.

For clarity, let $\mathcal{V}$ denote the learner set, and let $\mathcal{M}_{\text{si}}$ and $\mathcal{M}_{\text{st}}$ denote the simple and structured mechanism families. The category-level harm for task t and seed s is

$$\bar{X}_{t,s,c} = \frac{1}{|\mathcal{V}||\mathcal{M}_c||\mathcal{A}|} \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}_c} \sum_{a \in \mathcal{A}} X_{(t,m,a,s),t}, \quad (24)$$

so the prespecified contrast is

$$\hat{\Gamma}_{\text{si-st}} = \frac{1}{|\mathcal{T}||\mathcal{S}|} \sum_{t \in \mathcal{T}} \sum_{s \in \mathcal{S}} (\bar{X}_{t,s,\text{si}} - \bar{X}_{t,s,\text{st}}). \quad (25)$$

Writing the seed-averaged paired contrast as e_t , its exact two-sided task-level sign-flip value is

$$p_{\text{flip}} = 2^{-|\mathcal{T}|} \sum_{e \in \{-1, +1\}^{|\mathcal{T}|}} \mathbf{1} \left[\left| \frac{1}{|\mathcal{T}|} \sum_t e_t e_t \right| \geq \left| \frac{1}{|\mathcal{T}|} \sum_t e_t \right| \right]. \quad (26)$$

For core confidence intervals, each bootstrap draw samples 20 tasks with replacement and then samples five injection seeds with replacement inside each selected task. All strengths, mechanisms, and learners associated with a drawn task-seed pair remain together. This preserves the pairing that defines X_{qt} and prevents the 27,500 model cells from acting as independent replicates. The simple-minus-structured interval is the 2.5th and 97.5th percentiles of 20,000 hierarchical draws. Its exact sign-flip test first averages seeds within task, enumerates all 2^{20} task-sign assignments, and uses the two-sided absolute mean. Holm adjustment is applied jointly to the three declared family contrasts. The eleven mechanism tests form a different Holm family; their profiles remain descriptive regardless of adjusted value.

For every repair coordinate u , governance differences are first averaged within each of the 20 controlled tasks. With $\mathcal{M}$, $\mathcal{A}$, and $\mathcal{S}$ denoting mechanisms, strengths, and injection seeds, the task-level effect for policy r , contract c , and budget p is

$$d_{t,r,c,p}^{(u)} = \frac{1}{|\mathcal{M}||\mathcal{A}||\mathcal{S}|} \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}} \sum_{s \in \mathcal{S}} \Delta_{q,r,c,p}^{(u)}, \quad (27)$$

and the registry-level estimand is the equal-task average

$$\hat{\Delta}_{r,c,p}^{(u)} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d_{t,r,c,p}^{(u)}. \quad (28)$$

For a mechanism family h with member set $\mathcal{M}_h$, replacing $\mathcal{M}$ by $\mathcal{M}_h$ in Eq. 27 gives

$$\hat{\Delta}_{r,c,p,h}^{(u)} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d_{t,r,c,p,h}^{(u)}. \quad (29)$$

We assess these estimands with a paired 5,000-draw cluster bootstrap over tasks. For draw b , tasks are sampled with replacement

and the complete task effect is carried with each sampled index:

$$\hat{\Delta}_{r,c,p}^{*(b,u)} = \frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} d_{T_i^{*(b)}, r,c,p}^{(u)}, \quad T_i^{*(b)} \stackrel{\text{iid}}{\sim} \text{Unif}(\mathcal{T}). \quad (30)$$

We report the observed mean task difference, percentile interval, and the fraction of bootstrap draws above zero:

$$\text{CI}_{0.95} = [Q_{0.025}(\hat{\Delta}_{r,c,p}^{*(u)}), Q_{0.975}(\hat{\Delta}_{r,c,p}^{*(u)})],$$

$$\hat{P}_{\Delta^{(u)} > 0} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{\Delta}_{r,c,p}^{*(b,u)} > 0]. \quad (31)$$

The latter is a bootstrap superiority probability, not a p -value. Because tasks form a designed registry rather than a sample from a defined task population, all intervals quantify stability to registry reweighting. They do not support population inference to arbitrary tabular datasets.

4 Measurement Results

4.1 Simple Contamination Produces More Harm on Average

The prespecified simple-minus-structured contrast in paired AUROC harm was 0.164 (95% hierarchical bootstrap CI [0.150, 0.178]; Holm-adjusted exact task-level sign-flip $p = 5.72e - 06$). The direction was positive across all five core learner families. This category average does not imply that every structured mechanism is harmless: M09 is a structured counterexample with mean harm 0.235.

The learner-family consistency is descriptive rather than a second support decision. LR, bagged trees, boosted trees, and TabM differ in their ability to represent interactions and partitions, but each yielded a positive simple-minus-structured average. This observation reduces the likelihood that the pooled direction is produced by one learner family alone. It does not show that learner capacity causes the category contrast, because hyperparameters, regularization, and representation are fixed components of the named model implementations rather than randomized treatments.

The mechanism rows further qualify the category summary. M01 and M02 show the expected vulnerability to explicit target proxies, while redundant M06 and mixed M10 yield still larger mean harm. M04 and M05 are construction-invalid despite estimates close to zero: their future-window values can be weakly useful in this particular generator and representation. M09 demonstrates the opposite possibility. Its source categories are structurally defined, yet the complete one-hot representation makes the source exploitable. Therefore “structured” describes how invalid information is organized, not a guarantee of low statistical association or low downstream harm.

4.2 Detectability and Exploitability Form Crossed Profiles

Mechanism profiles show why a single leakage severity score is insufficient. M03 combines low primary-MI detectability ($D=0.079$) with substantial harm ($X=0.217$). M08 has intermediate detectability ($D=0.356$) and an imprecise small mean harm ($X=0.003$, CI spanning zero). M09 is both more detectable and strongly exploitable in the fixed one-hot source representation. These rows are descriptive profiles, not post hoc thresholded claims.

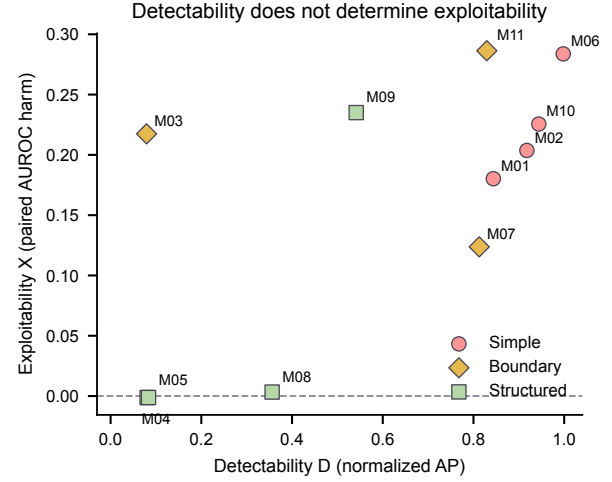


Figure 2: C/D/X measurement profiles. Each point is one mechanism mean over the frozen registry. D is normalized average precision under the primary training side mutual information diagnostic; X is paired AUROC harm. Shape and color encode construction family. The figure is descriptive and does not fit a deployment rule.

The diagnostic sensitivity suite makes the conditioning on D concrete. Across all mechanisms, the aggregate ordering of the four diagnostics differs from their ordering on individual mechanisms. M03 is the clearest example: its normalized AP ranges from 0.035 under absolute correlation to 0.602 under RF permutation importance, whereas primary MI yields 0.079. M04 and M05 remain weakly localized under all four evaluated diagnostics. These comparisons show that “hard to detect” must name a score family and representation. They do not establish RF permutation as a universally superior detector, because the method comparison lacks a frozen simultaneous paired interval and the best-per-mechanism choice uses oracle labels retrospectively.

At the eleven-mechanism level, D and X have a positive descriptive rank association, but category membership explains much of the apparent relation and leave-one-mechanism-out checks do not support a portable prediction rule. This is why Fig. 2 is a profile map rather than a fitted calibration curve. The benchmark can identify mechanisms for which a diagnostic and a learner co-vary; it cannot infer exploitability for an unseen mechanism from D alone.

4.3 Natural Cases Broaden Context but Not the Sampling Frame

Five selected real-data case studies provide boundary-specific consistency checks (Table 2): UCI Bank Marketing [17], a local Lending Club mirror, U.S. flight delays [18], Chicago food inspections [2], and NYC 311 [3]. All have positive mean paired harm, but their size, feature representation, and contamination labels differ. The mean over these five selected tasks is 0.273. We do not use these cases for population inference.

The five case means span a wide range rather than forming a homogeneous replication set. Flight and inspection boundaries show large observed harm, whereas the NYC 311 boundary is

Table 1: Measurement profiles for the frozen mechanism registry. Harm is permissive-minus-strict test AUROC. D is normalized average precision for the primary mutual-information ranker. Harm intervals are task-hierarchical 95% intervals. Mechanism rows are descriptive; the prespecified simple-minus-structured contrast is reported in the text.

ID	Mechanism	Family	Harm	95% CI	D
M01	Direct Target Copy	simple	+0.180	[+0.164, +0.196]	0.844
M02	Noisy Target Proxy	simple	+0.204	[+0.186, +0.222]	0.918
M03	Nonlinear Target Transform	boundary	+0.217	[+0.201, +0.234]	0.079
M04	Post-Outcome Aggregation	structured	-0.001	[-0.004, +0.001]	0.081
M05	Temporal Look-Ahead	structured	-0.001	[-0.003, +0.001]	0.084
M06	Redundant Leakage Cluster	simple	+0.284	[+0.261, +0.306]	0.998
M07	Sparse Subgroup Leakage	boundary	+0.124	[+0.114, +0.133]	0.813
M08	Entity Leakage	structured	+0.003	[-0.002, +0.009]	0.356
M09	Source Leakage	structured	+0.235	[+0.216, +0.254]	0.541
M10	Mixed Leakage	simple	+0.226	[+0.206, +0.245]	0.944
M11	Graph-Mediated Leakage	boundary	+0.286	[+0.263, +0.309]	0.829

close to the strict reference. This spread is substantively useful: it shows that declaring a field unavailable does not determine the magnitude of model exploitation. However, the tasks differ in sample construction, table width, encoded representation, unavailable-field count, and application semantics. Averaging them provides a compact audit summary, not a transport estimate. The fact that all five observed means are positive is reported as a fixed-case description and is not promoted to a population claim.

The matched-cost governance sensitivity is also heterogeneous. The descriptive mean P3-minus-mean-P2 SDR across the five cases is +0.187 (case-bootstrap interval [+0.025, +0.323]), but the five cases do not form a population sample. Bank Marketing (+0.102), Lending Club (+0.266), BTS Flights (+0.348), and Chicago Food (+0.326) favor P3, whereas NYC311 favors P2 (-0.108). The exact two-sided sign-flip value over the five fixed case effects is 0.1875. We therefore classify natural governance as mixed descriptive evidence rather than external confirmation. NYC311 also has only +0.013 initial harm. Its negative repair contrast is therefore consistent with a low-opportunity case in which deleting highly associated legitimate fields can cost more than the small removable distortion; it is not evidence for a general natural-data failure mode.

5 Governance Results

5.1 Positive Localization Does Not Pass the Repair Gate

Table 3 reports the primary semantic-group 20% contract. For P3, directional repair improves over the 20-seed matched-random mean by +0.095 (95% CI [+0.079, +0.110]), and complete leakage-group recall improves by +0.483 ([+0.462, +0.503]). These are substantive localization gains. However, P3 also increases overcorrection by +0.052 ([+0.023, +0.085]), while its legacy SDR interval crosses zero (+0.043, [-0.005, +0.084]). The joint CDXR status is therefore TRADEOFF, not successful repair.

P4–P6 reproduce the same qualitative conflict. Each improves direction and complete-group recall, and each reliably increases overcorrection. P6 achieves the largest full-group recall contrast (+0.583), but this does not dominate the other policies because it also incurs overcorrection (+0.059) and a larger legitimate-retention intervention. No scalar ranking is reported without a declared utility over these coordinates.

5.2 Budget Traces a

Localization–Overcorrection Frontier

P3’s semantic-group contrast changes systematically with budget. At 5%, SDR is +0.115 (CI [+0.098, +0.130]), directional repair is +0.120 ([+0.106, +0.133]), and overcorrection is +0.005 ([+0.001, +0.009]). At 10%, the corresponding values are +0.085, +0.109, and +0.024; at 20%, they become +0.043, +0.095, and +0.052. Complete-group recall remains large at all three fractions (+0.539, +0.537, and +0.483 relative to P2). Increasing budget therefore does not monotonically improve repair. It spends additional intervention capacity after many invalid groups are already localized, increasing the probability of crossing the strict reference.

This pattern also explains why we do not choose 5% retrospectively as the “winning” budget. The 5% point has the strongest SDR and smallest overcorrection, but the zero-tolerance semantic gate still detects a reliable adverse coordinate, and the budget was one member of a frozen three-point design. A deployment could prefer that point under a declared nonzero overcorrection tolerance. CDXR requires that tolerance to be specified as part of the utility or claim gate before selecting a result.

Mechanism decomposition shows that the frontier is not a family slogan. At 20%, P3 has positive directional effects for M01, M02, M06, M09, M10, and M11, but negative directional effects for M03–M05 and M08. M09 remains a structured counterexample (+0.175 direction; +0.107 SDR), while M04, M05, and M08 have negative SDR contrasts. Hence “structured leakage fails” is too coarse: construction category, repair opportunity, marginal localization, and group representation jointly determine the observed intervention.

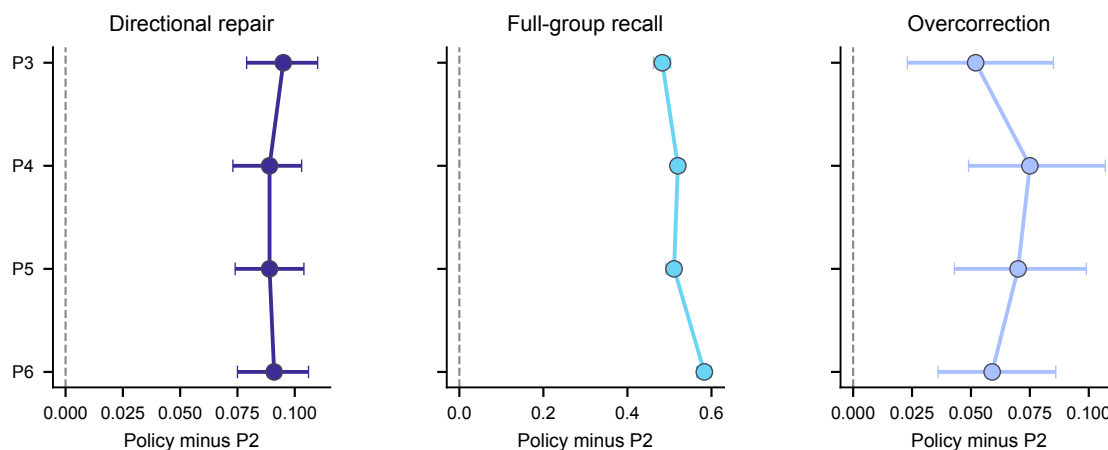
Pairwise policy comparisons are likewise metric-scoped. For semantic-group 20% SDR, P3 exceeds P4 by +0.030 (CI [+0.017, +0.045]) and P5 by +0.023 ([+0.011, +0.038]), but its contrast with P6 crosses zero (+0.011, [-0.003, +0.026]). Because P6 also has the largest full-group recall and positive overcorrection, these rows define alternatives on a frontier rather than a total order.

5.3 Random Variance and Cost Are Part of the Estimand

Random deletion is not a single convenient realization. Full-B1 evaluates 20 P2 seeds for every key, budget, and cost contract, then averages them before forming each paired policy contrast.

Table 2: Boundary-specific real-data case studies. These selected tasks provide descriptive consistency checks, not population inference.

Task	Declared prediction boundary	Features	Leak fields	Mean harm
BankMarketing	Before call completion	20	1	+0.131
LendingClub	At loan origination	140	29	+0.319
BTSFlights	Before scheduled departure	108	72	+0.470
ChicagoFood	Before inspection outcome	16	2	+0.430
NYC311	At complaint creation	40	2	+0.013

**Figure 3: Full-B1 repair-vector contrasts under semantic-group 20% cost. Points are learned-policy minus within-key mean P2 effects; bars are task-cluster 95% intervals. Directional repair and full-group recall favor the learned policies, but overcorrection rises simultaneously.**

Thus uncertainty over which fields a random policy happens to remove is integrated inside the estimand rather than mistaken for between-task uncertainty.

The cost contract changes semantic behavior more than score behavior. At 20%, semantic-group minus encoded-column full-group recall is +0.144 for P3 (CI [+0.138,+0.150]), whereas the corresponding legacy-SDR change is -0.001 ([-0.010,+0.007]). Semantic grouping prevents partial one-hot block removals and improves the construct alignment of localization, but it does not create a stronger score-recovery claim. This is precisely why κ belongs in the claim statement rather than in implementation footnotes.

5.4 Archetype and Learner Boundaries

The earlier sparse-regime failure is not reproduced under the formal primary contract. P3's sparse legacy-SDR contrast is +0.016 (CI [-0.070,+0.067]), its directional contrast is +0.077 ([+0.047,+0.106]), and overcorrection is +0.061 ([+0.022,+0.116]). This is neither a robust benefit nor a robust score failure; it is another repair tradeoff. Interaction tasks show the largest positive SDR contrast (+0.089), but each archetype contains only four designed tasks, so these rows are sensitivity analyses rather than independent environment-level inference.

The separately frozen learner amendment is narrower than Full-B1. At encoded-column 20%, the legacy P3-minus-P2 SDR contrasts are +0.043 for LR, +0.055 for RF, and +0.056 for LightGBM; direct interaction intervals cross zero. We therefore report that no reliable learner interaction was detected for this one score coordinate. We do not claim learner invariance: directional

repair exhibits a small LR disadvantage, and all three learners fail the overcorrection gate. Cross-learner consistency of score recovery cannot rescue a failed semantic-repair claim.

5.5 What CDXR Changes

Without CDXR, the encoded-cost P3 SDR interval could be summarized as a positive governance result. The architecture rejects that shortcut twice. First, semantic-group cost is the unit aligned with the M09 construction. Second, the repair vector reveals that better localization and directional movement coexist with reliable overcorrection. The defensible finding is not that MI repairs leakage, nor that structured leakage defeats MI. It is that blind association policies trace a reproducible localization-overcorrection frontier whose acceptability depends on an explicit downstream utility.

6 Related Work

Leakage semantics. Classical leakage formulations distinguish information that is available when a model is developed from information that will be available when its prediction is consumed [9]. This semantic perspective is stronger than a correlation rule: an aggressively predictive field may be legitimate, while a weak future-derived field remains invalid. Recent work links leakage to reproducibility in ML-based science [8] and formalizes domain-specific label leakage in health care [4]. Empirical studies in connectome prediction further show that leakage can inflate reported performance under otherwise familiar modeling pipelines [14]. LeakBench-Tab adopts the same

Table 3: Full-B1 policy contrasts under the primary semantic-group 20% contract. Each entry is the learned policy minus the within-key mean of 20 matched-random P2 seeds, with task-cluster bootstrap 95% intervals. Positive direction and full-group recall are desirable; positive overcorrection is a cost. The joint gate classifies every policy as a tradeoff.

Policy	Status	Δ SDR	Δ direction	Δ full recall	Δ overcorr.
P3	Tradeoff	+0.043 [-0.005,+0.084]	+0.095 [+0.079,+0.110]	+0.483 [+0.462,+0.503]	+0.052 [+0.023,+0.085]
P4	Tradeoff	+0.013 [-0.033,+0.053]	+0.089 [+0.073,+0.103]	+0.520 [+0.509,+0.530]	+0.075 [+0.049,+0.107]
P5	Tradeoff	+0.019 [-0.025,+0.057]	+0.089 [+0.074,+0.104]	+0.511 [+0.491,+0.527]	+0.070 [+0.043,+0.099]
P6	Tradeoff	+0.032 [-0.009,+0.068]	+0.091 [+0.075,+0.106]	+0.583 [+0.565,+0.600]	+0.059 [+0.036,+0.086]

boundary-first principle, but turns it into a factorial measurement contract in which validity, localization, and exploitation are separate variables.

Detection at code, data, and representation levels. Static program analysis targets leakage pathways visible in data-science code, such as fitting preprocessing outside the training split [16]. Contamination audits in language benchmarks instead often ask whether evaluation content was memorized and subsequently exploited [11]. Feature-level screening occupies another level: it ranks columns from observed training data but typically cannot determine whether their information is temporally or operationally available. These approaches are complementary because they observe different evidence. Code analysis can reveal an invalid fit scope without ranking columns; a statistical ranker can reveal a target proxy without knowing its lineage; and a semantic contract can identify a future field even when its sample association is weak. Our CDXR architecture makes these non-equivalences explicit rather than combining them into a single detector score.

Tabular evaluation under controlled shifts. TableShift studies tabular prediction under distribution shifts, while broad comparative benchmarks examine when boosted trees, bagged trees, linear models, and neural architectures succeed [5, 7, 12]. Those resources vary datasets, domains, or learner families to characterize predictive performance. LeakBench-Tab holds a task instance and split fixed while perturbing the information boundary through auditable mechanisms. Learner diversity is used to measure exploitability conditional on representation, not to declare a new winner among logistic regression, random forests, boosting, and TabM [1, 6, 10, 13]. This focus also explains why the natural cases are treated as boundary audits rather than merged with the controlled panel.

Feature selection versus leakage governance. Mutual information, linear coefficients, correlation, and permutation importance measure predictive relevance under different function classes. None is intrinsically a semantic validity test. Conventional feature selection usually optimizes predictive performance or sparsity; leakage governance asks whether a constrained intervention moves an evaluation toward a boundary-valid reference while preserving legitimate information. Our P2–P6 comparison therefore matches declared cost, records complete-group localization and legitimate retention, decomposes movement into direction and overcorrection, and applies a falsification gate. The technical contribution is the reusable evaluation interface and its empirical instantiation, not a new ranker.

7 Discussion

7.1 What the Evaluation Changes

Leakage audits often end after declaring suspicious fields or measuring an inflated score. Our results show that an actionable evaluation needs at least four linked but distinct objects: the semantic contamination label, a blind localization score, downstream exploitability, and the effect of a removal policy under explicit cost. Reporting only one object hides important cases. Low D does not validate a field, high X does not reveal which field caused the inflation, and high D does not guarantee that removal will preserve legitimate utility.

Matched-cost evaluation is central to this conclusion. Each learned policy is compared with 20 random selections of the same charged cost, under both encoded and semantic units. Even that comparison is insufficient by itself: positive SDR can arise after crossing the strict reference or deleting the wrong fields. The direction, group-recall, retention, and overcorrection coordinates expose these paths.

7.2 Implications for Leakage Audits

The findings suggest an audit sequence rather than an automatic deletion rule. An analyst should first declare the prediction event and label each field by availability or lineage. Blind scores can then prioritize manual review, but their rankings should be evaluated against a held-back semantic mask when a benchmark is available. Finally, any removal policy should be compared with an equal-cost negative control and assessed on downstream behavior. This sequence prevents a high association score from silently becoming a validity judgment.

Budget and policy choice are governance decisions, not benchmark winners. A deployment process needs a domain-specific utility over missed contamination, lost legitimate signal, calibration, and decision consequences. Full-B1 deliberately preserves the vector outcome so a future user can declare that utility without rerunning selection or retroactively choosing a favorable metric.

The cost comparison shows why representation must be part of the audit contract. Grouping M09’s indicators strongly improves full-group recall while barely changing the aggregate SDR contrast. Richer policies can add lineage, temporal, or entity metadata through $\mathcal{A}$, without changing the evaluator. The hidden contamination label must remain an evaluation object.

7.3 Alternative Explanations and Required Follow-Ups

The observed family pattern can have several compatible explanations. Simple mechanisms may be easier for MI to rank because their invalid signal is carried by individual encoded

columns. Structured mechanisms may require group or conditional statistics, but they may also exhibit weaker exploitability in the particular generators, leaving less inflation for any policy to repair. Learner representation adds a third possibility, although the similar LR, RF, and LightGBM contrasts do not reveal a detectable learner interaction in the present panel. The current factorial design describes the joint outcome of these factors; it does not identify their separate causal contributions.

The largest remaining ambiguity concerns policies that use lineage, temporal, or entity metadata rather than blind univariate association. These policies would test whether low-gap structured mechanisms become governable when the intervention represents their construction semantics. Natural evaluation also remains narrow: five fixed cases cannot establish a population-level transport claim, and the negative NYC311 effect requires application-specific explanation rather than averaging away.

7.4 Limitations

The controlled tasks are synthetic panel generators, not twenty independent real-world datasets. Their bootstrap intervals describe registry stability, not a task population. Full-B1 uses LR downstream. The separately frozen P3 amendment covers LR, RF, and LightGBM only at encoded-column 20%; those named implementations do not establish invariance across model families, policies, cost contracts, or tuning choices. Interaction intervals crossing zero are absence of detected difference, not equivalence.

P2 uses 20 frozen random selections for each key, budget, and cost contract, and the primary estimand averages P2 within key before task-cluster resampling. This integrates the declared random-policy variance but does not cover every possible randomization scheme. Nominal budget labels can also map to the same integer removal count on tasks with few fields, so adjacent budget points are not fully distinct interventions for every task.

All four learned policies are association-based. None encodes lineage, temporal availability, entity semantics, or source provenance. Their tradeoff statuses should not be generalized to metadata-aware policies. Conversely, positive localization must not be read as evidence that association determines validity.

The five real-data audits are selected, boundary-specific case studies. One local Lending Club mirror cannot be redistributed until its source and license are verified. AUROC distance does not replace calibration, decision utility, or operational monitoring. Runtime and resource overhead are also outside the current frozen evidence and should be added before making systems-efficiency claims.

7.5 Threats to Validity

Construct validity. SDR measures movement toward a frozen strict reference, not the business value of a deployed decision. Its interpretation depends on the correctness of each declared prediction boundary and mechanism contract. Recall and retention use the contamination mask only after policy selection and are therefore evaluation measures rather than deployable observables. Complete-group recall and both cost contracts address one-hot expansion, but the quality of the semantic grouping remains part of the information contract.

Internal validity. The paired protocol holds splits, task content, model seeds, and removal cost fixed within each comparison. Bundle, split, and runner hashes reduce implementation

drift, but they cannot rule out all library-specific behavior. Hyperparameters are fixed named implementations rather than optimized independently for every mechanism; this improves comparability but may understate the response of a specially tuned learner. Protocol amendments are provenance-bound and replace affected cells, yet their necessity also illustrates how semantic bugs can survive ordinary completion checks.

External and statistical validity. The designed registry deliberately varies mechanism structure, but it does not define a sampling frame of tabular applications. Mechanism-family intervals and bootstrap probabilities should be read as stability summaries for this registry. The natural cases broaden the contexts inspected but remain selected case studies; their governance result is mixed and descriptive. Multiplicity correction protects the declared contrast families, but descriptive mechanism profiles and diagnostic sensitivity rows should not be mined as independent confirmatory discoveries.

8 Reproducibility and Artifact Boundary

The release binds the canonical 27,500-cell measurement table, the 803,000-cell Full-B1 merge, validation receipt, repair claim state, bootstrap analysis, and paper-facing tables by SHA-256. The Full-B1 suite reports 312 passing tests with no failures. Superseded runners and per-shard execution cache are excluded from claim generation; the canonical merge and deterministic summaries are tracked.

Reproduction is organized in three layers. The *paper layer* regenerates all numeric macros, tables, and figures from compact governed CSVs and the Full-B1 claim tables, and records their hashes in a paper-artifact manifest. The *analysis layer* rebuilds measurement and governance summaries from canonical cell ledgers, including the task-level bootstrap objects. The *execution layer* reruns frozen task bundles through named model and policy runners after verifying bundle, split, configuration, and code identities. A reader can therefore inspect the paper without re-running models, recompute claims without rebuilding tasks, or execute the complete experiment when the required packages and hardware are available.

Claim state is machine-readable and distinct from prose. Each central claim records an estimand, evidence tier, allowed wording, prohibited wording, and scope limitation. The paper generator does not infer support from arbitrary tables; it consumes the frozen paper-facing assets. This separation is intended to prevent a later figure edit, exploratory table, or superseded run from silently changing a conclusion.

The public artifact and reproduction instructions are maintained at <https://github.com/lzzzh/LeakBench-Tab>. The paper-facing artifact does not redistribute the local-only Lending Club mirror. All non-oracle policy selectors can be audited without access to the contamination mask; the mask is used only for evaluation after selection.

9 Conclusion

CDXR turns leakage repair from an association ranking into a falsifiable evaluation contract. Its interfaces specify prediction-time information, policy-visible metadata, intervention cost, oracle isolation, paired outcomes, and the conditions under which a repair claim is unsupported. LeakBench-Tab instantiates the architecture at measurement and governance scale.

The conclusion is narrower than the legacy SDR result. P3–P6 improve complete group localization and directional movement

over 20-seed matched random removal, but all increase overcorrection; their status is tradeoff. Cross-learner score consistency does not remove that conflict. Future metadata-aware policies can use the same CDXR interface and should be accepted only when they improve the joint repair vector rather than one convenient coordinate.

References

- [1] Leo Breiman. 2001. Random Forests. *Machine Learning* 45 (2001), 5–32. doi:10.1023/A:1010933404324
- [2] City of Chicago. 2026. Food Inspections. Chicago Data Portal. <https://data.cityofchicago.org/d/4ijn-s7e5> Accessed 2026-07-13.
- [3] City of New York. 2026. 311 Service Requests from 2020 to Present. NYC Open Data. <https://data.cityofnewyork.us/d/erm2-nwe9> Accessed 2026-07-13.
- [4] Sharon E. Davis, Michael E. Matheny, Suresh Balu, and Mark P. Sendak. 2024. A Framework for Understanding Label Leakage in Machine Learning for Health Care. *Journal of the American Medical Informatics Association* 31, 1 (2024), 274–280. doi:10.1093/jamia/ocad178
- [5] Josh Gardner, Zoran Popovic, and Ludwig Schmidt. 2023. Benchmarking Distribution Shift in Tabular Data with TableShift. In *Advances in Neural Information Processing Systems*, Vol. 36.
- [6] Yury Gorishniy, Akim Kotelnikov, and Artem Babenko. 2025. TabM: Advancing Tabular Deep Learning with Parameter-Efficient Ensembling. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=Sd4wYYOhmY>
- [7] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. 2022. Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?. In *Advances in Neural Information Processing Systems*, Vol. 35.
- [8] Sayash Kapoor and Arvind Narayanan. 2023. Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. *Patterns* 4, 9 (2023), 100804. doi:10.1016/j.patter.2023.100804
- [9] Shachar Kaufman, Saharon Rosset, and Claudia Perlich. 2011. Leakage in Data Mining: Formulation, Detection, and Avoidance. In *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 556–563. doi:10.1145/2020408.2020496
- [10] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems*, Vol. 30.
- [11] Inbal Magar and Roy Schwartz. 2022. Data Contamination: From Memorization to Exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. 157–165. doi:10.18653/v1/2022.acl-short.18
- [12] Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Vishak Prasad C, Ganesh Ramakrishnan, Micah Goldblum, and Colin White. 2023. When Do Neural Nets Outperform Boosted Trees on Tabular Data?. In *Advances in Neural Information Processing Systems*, Vol. 36.
- [13] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2018. CatBoost: Unbiased Boosting with Categorical Features. In *Advances in Neural Information Processing Systems*, Vol. 31.
- [14] Matthew Rosenblatt, Link Tejavibulya, Rongtao Jiang, Stephanie Noble, and Dustin Scheinost. 2024. Data Leakage Inflates Prediction Performance in Connectome-Based Machine Learning Models. *Nature Communications* 15 (2024), 1829. doi:10.1038/s41467-024-46150-w
- [15] Takaya Saito and Marc Rehmsmeier. 2015. The Precision–Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE* 10, 3 (2015), e0118432. doi:10.1371/journal.pone.0118432
- [16] Pavle Subotić, Uroš Bojanić, and Milan Stojić. 2022. Statically Detecting Data Leakages in Data Science Code. In *Proceedings of the 11th ACM SIGPLAN International Workshop on the State Of the Art in Program Analysis*. 16–22. doi:10.1145/3520313.3534657
- [17] UCI Machine Learning Repository. 2014. Bank Marketing. doi:10.24432/C5K306 CC BY 4.0.
- [18] U.S. Bureau of Transportation Statistics. 2026. Reporting Carrier On-Time Performance Data. TranStats data portal. https://www.transtats.bts.gov/DL_SelectFields.aspx?QO_fu146_anzr=&gnoyr_VQ=FGJ Accessed 2026-07-13.

1198
1199
1200
1201
1202
1203
1204
1205
1206
1207
1208
1209
1210
1211
1212
1213
1214
1215
1216
1217
1218
1219
1220
1221
1222
1223
1224
1225
1226
1227
1228
1229
1230
1231
1232
1233
1234
1235
1236
1237
1238
1239
1240
1241
1242
1243
1244
1245
1246
1247
1248
1249
1250
1251
1252
1253
1254
1255
1256
1257
1258
1259
1260